

ABHISHEK VASANTKUMAR JANI

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EDUCATION

Rutgers University

Sep 2024 - May 2026

Master of Science, Computer Science (GPA: 3.83 out of 4)

- **Coursework:** Machine Learning, Natural Language Processing, Computer Vision, Deep Learning, Database Systems, Data Structures & Algorithms, Interpretable AI

Ganpat University

Jul 2019 - May 2023

Bachelor of Technology, Computer Engineering (GPA: 8.97 out of 10)

SKILLS

- **Programming Languages:** Python, C++, Java, Javascript, TypeScript, C
- **Web & Frameworks :** Node.js, Express, React, Next.js, Flask, Django, FastAPI
- **Databases:** PostgreSQL, MongoDB, MySQL, SQLite, ChromaDB
- **ML/Data:** TensorFlow, PyTorch, Scikit-Learn, Transformers (BioBERT), MediaPipe, NumPy, Pandas, Matplotlib, OpenCV
- **Cloud & DevOps:** AWS (Lambda, Step Functions, EventBridge, CDK, IAM), Docker, Prometheus, Grafana
- **Tools:** Git, Linux, Restful API, OAuth 2.0

EXPERIENCE

Attention Group LLC | *Software Engineering Intern*

Jun 2025 - Dec 2025

- Shipped production Node.js/Express + MongoDB backend with 30+ REST endpoints, implementing authentication middleware, rate limiting, and error handling serving 2K+ requests/minute at 99.9% uptime
- Built the image-generation workflow-including prompt templating, captioning, overlay composition, pin/persist, and metadata-using React hooks, OpenCV for image processing, and Matplotlib for visual previews, which improved user experience and workflow efficiency
- Implemented OAuth 2.0 authentication system for Twitter API integration, building secure token management and callback handling that reduced login errors by 85% (from 12% to <2% failure rate)

Sophos | *Technical Support Engineer - L1*

Aug 2023 - May 2024

- Resolved 200+ technical incidents for critical network security modules (IPSec, SSL VPN, VoIP, firmware failover) using systematic root-cause analysis and proactive troubleshooting, reducing average incident resolution time by 18%
- Optimized support workflows utilizing Salesforce CRM, reducing average first-response time by 15% through strategic queue prioritization and real-time escalation protocols
- Documented 100+ technical solutions and troubleshooting workflows in internal knowledge base, achieving 95% customer satisfaction rating and reducing escalation rate to L2 by 25%

Ganpat University | *Research Intern*

Jan 2023 - May 2023

- Improved image classification accuracy by 25% on EMOTIC dataset by engineering hybrid CNN model combining AlexNet and VGG19, leveraging transfer learning and advanced preprocessing techniques
- Conducted a comparative evaluation of optimizers on ensemble models for emotion recognition, achieving 73.6% validation accuracy and informing hyperparameter tuning strategies

PROJECTS

Data Reliability Guardrails | [Github](#)

- Developed a fault-tolerant validation system that enforces strict data contracts (Schema, Freshness, Distribution), preventing silent data failures in batch pipelines.
- Implemented the Population Stability Index (PSI) algorithm and volume checks to automatically detect distribution shifts, gating downstream consumers with <5ms latency.
- Built a custom benchmarking suite with failure injection (simulating schema drift, latency, and spikes) to verify system stability, achieving a 98% automated detection rate.

Runbook Ranger (AWS Auto-Remediation Bot) | [Github](#)

- Engineered a serverless self-healing system using AWS Lambda, Step Functions, and EventBridge to detect and resolve incidents automatically, reducing MTTR from minutes to seconds.
- Implemented production-grade safety mechanisms, including distributed resource locking, idempotency keys, and blast-radius limits, ensuring automation never causes outages.
- Built a Runbooks-as-Code engine (Python/YAML) that treats remediation logic as versioned software, deployed via AWS CDK with strict IAM least-privilege policies.

Extending ADE Detection with Explainable Counterfactuals and BioBERT | [Github](#)

- Fine-tuned BioBERT (dmis-lab/biobert-base-cased-v1.1) on ADE-Corpus V2 dataset using Hugging Face Transformers over 10 epochs, achieving 95.6% F1-score (95.3% precision, 96.2% recall) in adverse drug event classification across 11,914 clinical sentences
- Implemented counterfactual analysis by masking drug entities across 6,821 ADE-positive and 5,093 non-ADE sentences, revealing 76.5% prediction flip rate in ADE cases vs. 1.9% in non-ADE cases, quantifying the model's reliance on drug entity features
- Evaluated lexical robustness by substituting 541 generic drug names with brand names using RxNorm API across 8,448 sentences, identifying 13.1% prediction instability in ADE-positive cases and demonstrating areas for model hardening in clinical deployment

HONORS & ACTIVITIES

Amazon | *Selected Participant, Amazon ML Summer School*

Jul 2022

- Selected for an intensive virtual learning program focused on cutting-edge Machine Learning topics, including Supervised Learning, Deep Neural Networks, Dimensionality Reduction, Unsupervised Learning, Probabilistic Graphical Models, Sequential Learning, Causal Inference, and Reinforcement Learning

Sophos | *Sophos Certified Engineer*

Sep 2023

- Attained certification in Sophos Firewall v19.5, v20.0 and MDR, demonstrating expertise in network security protocols